

Some interesting features

- Simulation is possible in single, di and tri dimensions as well as in cylindrical coordinates
- Effective exploitation of symmetries including odd/even mirror symmetries and 90°/180° rotations allow reduction of computation size
- Field output provided in Hierarchical Data Format (HDF5) is supported by many visualisation tools
- Complete scriptability provided through Scheme scripting front-end or a callable C++ library or python binding
- Field analysis including frequency extraction, near- to far-field transformation, flux spectra, etc is fully programmable

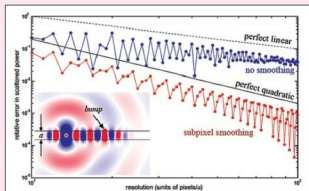


Fig. 2: Plot showing the relative error in scattered power from a small semi-circular bump in a dielectric waveguide excited by a point-dipole source in the waveguide, as a function of the computational resolution. The geometry and field of the waveguide is shown in the inset

for an arbitrary source such as a continuous wave input.

Many other quantities could be easily computed with the help of these results. Various computations could be added either sequentially or in parallel with the help of the scriptable interface.

Other competing software

Let us now take a look at some other FDTD implementations available as open source.

Angora. Main features of this free FDTD implementation include an MPI library for full parallelisability, build and install mechanism for GNU/Linux operating system being automatic, support for planar multi-layered spaces, flexible and user-friendly configuration with the help of text based configuration files and numerical optical-imaging simulation.

OpenEMS. OpenEMS FDTD implementation comes with an extensive MATLAB/Octave interface. The software also has a graphical interface (GI) called CSXCAD. It can also

be effectively used for computation of far-field.

FDTD++. This is yet another actively-developed, fully-featured open source FDTD implementation. It provides flexible geometrical modelling capabilities with support for computer aided designs, meshes and algorithmic specifications.

GSVIT. This open source FDTD package is very helpful for scientific researches in nano technology and nano-scale optics. The tool supports the use of NVIDIA CUDA environment-compatible graphics cards. The latest version of the package comes along with a graphical interface, XSVit, for parameter files set up and visualisation purposes.

B-CALM. B-CALM, or Belgium California Light Machine, is an FDTD simulation tool that makes use of Compute Unified Device Architecture (CUDA) for fast computing on graphics card.

Dedicated EM simulation tool without a GI

There is a basic Unix philosophy that reads, 'Write programs that do one thing and do it well' and Meep software does exactly that. Developers visualise Meep as a dedicated tool for EM simulations, incorporating other functionalities by combining with other programs using standard interfaces.

Some users might find using the

Python Meep

The default scripting language for Meep software is Scheme. Python Meep is a wrapper around C++ core of Meep, allowing researchers to script Meep simulations using Python. Not only does it make the work of researchers easier, it also allows them to integrate Meep with other packages available in the Python ecosystem.

tool a bit difficult without a dedicated GI. "Setting up a realistic simulation is usually a challenging scientific task on its own and with Meep, one also needs to write a working code. That said, I believe that the minimalistic interface of Meep and the fundamental simplicity of the FDTD algorithm provide a very useful complement to more elaborate software packages," points out Filip Dominic in his blog titled 'MEEP - my notes on FDTD simulation.'

Developers have valid reasons to opt against a GI. Graphical simulations are used while setting up a single simulation. But this tool is not intended to have a single simulation. Simulations are to be done for a class of related problems and optimising the results and looking at the same geometry under a sequence of different stimuli. When it comes to a series of related simulations, a graphical computer aided design (CAD) style interface would not be optimal.

Second, developers do not want to write a mediocre CAD program. In the paper titled 'MEEP: A flexible free software package for electromagnetic simulations by the FDTD method,' they write, "If we wanted to use a CAD program, we would use a professional quality one, export the design to a standard interchange format and write a conversion program to turn this format into what Meep expects."

Therefore developers have finally come up with a tool that provides interactions using a low-level C++ interface, standard high-level scripting language called Scheme implemented by an external GNU Guile library and Python Meep. ●

The author is an electronics enthusiast from Kerala